Math 49

Precalculus with Limits

Math 49 Course at a Glance

Precalculus with Limits

TEXTBOOK | OpenStax Precalculus 2e (Jay Abramson, 2021)

Review Content

REVIEW	SECTIONS 1.2, 1.5, 4.1, 4.2, 4.3, 4.4, 4.5, 4.7
M49-FND-001	I can determine the domain and range of a function from a formula, graph, or table.
M49-FND-002	I can describe and apply transformations of a parent function.
M49-FND-004	I can analyze exponential functions using their equations, graphs, and contexts.
M49-FND-005	I can analyze logarithmic functions using their equations, graphs, and contexts.
M49-FND-006	I can solve exponential and logarithmic equations using equivalent forms and logarithm properties.

Function Operations, Composition, and Inverses

REQUIRED	SECTIONS 1.4, 1.7
M49-FND-003	I can determine whether a function is even, odd, or neither.
M49-OPS-001	I can add, subtract, multiply, and divide functions and state the resulting domain.
M49-OPS-002	I can evaluate a composition from formulas, graphs, or tables.
M49-OPS-003	I can write a formula for the composition of two functions and determine its domain.
M49-OPS-004	I can decompose a function into a composition of simpler functions.
M49-OPS-005	I can determine whether a function has an inverse on a given domain.
M49-OPS-006	I can find and verify an inverse function algebraically.
M49-OPS-007	I can interpret the relationship between a function and its inverse from graphs or tables.

Power, Polynomial, and Rational Functions

REQUIRED	SECTIONS 3.3, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, 4.2, 4.4
M49-PWR-001	I can identify a power function and interpret the parameters in y = kx^p.
M49-PWR-002	I can construct a power function from two data points or a proportional relationship.
M49-PWR-003	I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.
M49-PWR-004	I can recognize and model direct, inverse, and other power-variation relationships.
M49-PWR-005	I can solve equations involving integer or rational powers.
M49-PWR-006	I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.
M49-PWR-008	I can analyze power functions with fractional exponents, including their domains, graph shapes, and behavior near zero.
M49-ALG-002	I can predict the end behavior of a polynomial from its degree and leading coefficient.
M49-ALG-003	I can determine polynomial zeros and their multiplicities.
M49-ALG-004	I can construct a polynomial function from specified zeros or graphical features.
M49-ALG-005	I can sketch and analyze a polynomial function using intercepts, multiplicity, and end behavior.
M49-ALG-006	I can identify holes and vertical, horizontal, or slant asymptotes of a rational function.
M49-ALG-007	I can sketch and analyze a rational function using its algebraic and asymptotic features.
M49-ALG-008	I can build or select a power, polynomial, or rational model for a context.

Limits and Function Comparison

REQUIRED	SECTIONS 12.1
M49-LIM-001	I can interpret limit notation as a statement about function behavior near an input.
M49-LIM-002	I can estimate a limit from a graph or table.
M49-LIM-003	I can evaluate a limit algebraically using direct substitution or simplification.
M49-LIM-005	I can interpret an infinite limit as vertical-asymptotic behavior.
M49-LIM-006	I can determine limits at infinity and relate them to long-run behavior.
M49-LIM-007	I can compare the long-run growth of polynomial, exponential, logarithmic, and rational functions.

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Sequences and Series

REQUIRED	SECTIONS 11.1, 11.2, 11.3, 11.4
M49-SEQ-001	I can evaluate and graph a sequence defined explicitly or recursively.
M49-SEQ-002	I can write explicit and recursive formulas for arithmetic sequences.
M49-SEQ-003	I can write explicit and recursive formulas for geometric sequences.
M49-SEQ-004	I can distinguish arithmetic, geometric, and other sequences from formulas, tables, or contexts.
M49-SEQ-005	I can create and interpret a sequence model for discrete change.
M49-SER-001	I can interpret and use summation notation.
M49-SER-002	I can calculate a finite arithmetic series.
M49-SER-003	I can calculate a finite geometric series.
M49-SER-004	I can determine whether an infinite geometric series converges and find its sum when it does.
M49-SER-005	I can create and interpret a series model for accumulated change.

Additional Limit Concepts

EXTENSION	SECTIONS 12.1
M49-LIM-004	I can determine one-sided limits and decide whether a two-sided limit exists.
M49-LIM-008	I can distinguish a function value from a limit and determine continuity at a point.

Parametric Equations

EXTENSION	SECTIONS 8.6, 8.7
M49-PAR-001	I can graph a plane curve defined by parametric equations with its orientation.
M49-PAR-002	I can eliminate a parameter to obtain a rectangular equation.
M49-PAR-003	I can create or interpret parametric equations for motion in the plane.

Conic Sections

EXTENSION	SECTIONS 10.1, 10.2, 10.3, 10.5
M49-CON-001	I can classify a circle, ellipse, or hyperbola from its equation or defining geometric relationship.
M49-CON-002	I can use completing the square to rewrite a general-form conic equation in standard form.
M49-CON-003	I can write, interpret, and graph implicit and parametric equations of a circle.
M49-CON-004	I can write, interpret, and graph implicit and parametric equations of an ellipse.
M49-CON-005	I can write, interpret, and graph implicit and parametric equations of a hyperbola.
M49-CON-006	I can determine the center, vertices, foci, axes, or asymptotes needed to describe a circle, ellipse, or hyperbola.
M49-CON-007	I can explain the geometric definitions of circles, ellipses, and hyperbolas using foci and distance relationships.
M49-CON-008	I can explain a parabola as the set of points equidistant from a focus and directrix.

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SKILL PROGRESSION

I	Introduce
R	Reinforce
D	Deepen
A	Apply

Review Content

REVIEW

UNIT TEXTBOOK COVERAGE
1.2 Domain and Range; 1.5 Transformation of Functions; 4.1 Exponential Functions; 4.2 Graphs of Exponential Functions; 4.3 Logarithmic Functions; 4.4 Graphs of Logarithmic Functions; 4.5 Logarithmic Properties; 4.7 Exponential and Logarithmic Models

M49-FND-001 — I can determine the domain and range of a function from a formula, graph, or table.

TEXTBOOK SECTIONS 1.2 Domain and Range

SUPPORTING SKILLS

D	Determine domain restrictions from denominators, even roots, and logarithms. (first introduced in Math 31) SK-FUNC-DOMAIN-FORMULA · Deepen
R	Read domain and range from a graph. (first introduced in Math 12) SK-FUNC-DOMAIN-RANGE-GRAPH · Reinforce
R	Read domain and range from a table. (first introduced in Math 12) SK-FUNC-DOMAIN-RANGE-TABLE · Reinforce

M49-FND-002 — I can describe and apply transformations of a parent function.

TEXTBOOK SECTIONS 1.5 Transformation of Functions

SUPPORTING SKILLS

R	Interpret transformations applied to function inputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-HORIZONTAL · Reinforce
R	Distinguish stretches from compressions and horizontal effects from vertical effects. (first introduced in Math 31) SK-FUNC-TRANSFORM-SCALE · Reinforce
R	Interpret transformations applied to function outputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-VERTICAL · Reinforce

M49-FND-004 — I can analyze exponential functions using their equations, graphs, and contexts.

TEXTBOOK SECTIONS 4.1 Exponential Functions; 4.2 Graphs of Exponential Functions; 4.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

R	Identify horizontal asymptotic behavior of an exponential graph. (first introduced in Math 31) SK-EXP-ASYMPTOTE · Reinforce
R	Identify initial amount, rate, and time interval in context. (first introduced in Math 31) SK-EXP-MODEL-CONTEXT · Reinforce
A	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Apply

M49-FND-005 — I can analyze logarithmic functions using their equations, graphs, and contexts.

TEXTBOOK SECTIONS 4.3 Logarithmic Functions; 4.4 Graphs of Logarithmic Functions; 4.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

R	Identify vertical asymptotic behavior of a logarithmic graph. (first introduced in Math 31) SK-LOG-ASYMPTOTE · Reinforce
R	Interpret a logarithm as an exponent. (first introduced in Math 31) SK-LOG-DEFINITION · Reinforce
A	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Apply

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SKILL PROGRESSION

I Introduce | R Reinforce | D Deepen | A Apply

M49-FND-006

I can solve exponential and logarithmic equations using equivalent forms and logarithm properties.

TEXTBOOK SECTIONS 4.3 Logarithmic Functions; 4.5 Logarithmic Properties; 4.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

R	Rewrite exponential expressions using a common base when possible. (first introduced in Math 31) SK-EXP-REWRITE-BASE · Reinforce
A	Reject solutions that make a logarithm argument nonpositive. (first introduced in Math 31) SK-LOG-DOMAIN-CHECK · Apply
A	Apply the logarithm power property in either direction. (first introduced in Math 31) SK-LOG-POWER · Apply
A	Apply the logarithm product property in either direction. (first introduced in Math 31) SK-LOG-PRODUCT · Apply
A	Apply the logarithm quotient property in either direction. (first introduced in Math 31) SK-LOG-QUOTIENT · Apply
R	Isolate an exponential expression and apply a logarithm. (first introduced in Math 31) SK-LOG-SOLVE-EXP · Reinforce
R	Combine or isolate logarithms before converting to exponential form. (first introduced in Math 31) SK-LOG-SOLVE-LOG · Reinforce

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SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Function Operations, Composition, and Inverses

REQUIRED

UNIT TEXTBOOK COVERAGE
1.4 Composition of Functions; 1.7 Inverse Functions

M49-FND-003 — I can determine whether a function is even, odd, or neither.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

D	Test a formula or graph for even or odd symmetry. (first introduced in Math 31) SK-FUNC-SYMMETRY · Deepen

M49-OPS-001 — I can add, subtract, multiply, and divide functions and state the resulting domain.

TEXTBOOK SECTIONS 1.4 Composition of Functions

SUPPORTING SKILLS

A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Determine the domain of a sum, difference, product, quotient, or composition. SK-FUNC-DOMAIN-COMBINE · Introduce
I	Add, subtract, multiply, and divide function outputs. SK-FUNC-OPERATIONS · Introduce

M49-OPS-002 — I can evaluate a composition from formulas, graphs, or tables.

TEXTBOOK SECTIONS 1.4 Composition of Functions

SUPPORTING SKILLS

D	Substitute one function formula into another. (first introduced in Math 31) SK-FUNC-COMPOSE-FORMULA · Deepen
D	Follow intermediate values to compose functions from tables. (first introduced in Math 31) SK-FUNC-COMPOSE-TABLE · Deepen
A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply

M49-OPS-003 — I can write a formula for the composition of two functions and determine its domain.

TEXTBOOK SECTIONS 1.4 Composition of Functions

SUPPORTING SKILLS

D	Substitute one function formula into another. (first introduced in Math 31) SK-FUNC-COMPOSE-FORMULA · Deepen
D	Determine the domain of a sum, difference, product, quotient, or composition. (first introduced in Math 39) SK-FUNC-DOMAIN-COMBINE · Deepen

M49-OPS-004 — I can decompose a function into a composition of simpler functions.

TEXTBOOK SECTIONS 1.4 Composition of Functions

SUPPORTING SKILLS

A	Substitute one function formula into another. (first introduced in Math 31) SK-FUNC-COMPOSE-FORMULA · Apply
I	Identify simpler inner and outer functions in a composition. SK-FUNC-DECOMPOSE · Introduce

M49-OPS-005 — I can determine whether a function has an inverse on a given domain.

TEXTBOOK SECTIONS 1.7 Inverse Functions

SUPPORTING SKILLS

D	Apply the horizontal line test for invertibility. (first introduced in Math 31) SK-FUNC-INVERSE-HLT · Deepen
R	Test whether a function is one-to-one on a domain. (first introduced in Math 31) SK-FUNC-ONE-TO-ONE · Reinforce

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SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M49-OPS-006 — I can find and verify an inverse function algebraically.

TEXTBOOK SECTIONS 1.7 Inverse Functions

SUPPORTING SKILLS

A	Substitute one function formula into another. (first introduced in Math 31) SK-FUNC-COMPOSE-FORMULA · Apply
D	Exchange variables and solve to obtain an inverse formula. (first introduced in Math 31) SK-FUNC-INVERSE-ALGEBRA · Deepen
D	Verify inverse functions using composition. (first introduced in Math 31) SK-FUNC-INVERSE-VERIFY · Deepen

M49-OPS-007 — I can interpret the relationship between a function and its inverse from graphs or tables.

TEXTBOOK SECTIONS 1.7 Inverse Functions

SUPPORTING SKILLS

D	Exchange domain and range between a function and its inverse. (first introduced in Math 31) SK-FUNC-INVERSE-DOMAIN-RANGE · Deepen
D	Exchange input and output values when interpreting an inverse. (first introduced in Math 31) SK-FUNC-INVERSE-SWAP · Deepen
D	Reverse input and output units when interpreting an inverse. (first introduced in Math 31) SK-FUNC-INVERSE-UNITS · Deepen

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Power, Polynomial, and Rational Functions

REQUIRED

UNIT TEXTBOOK COVERAGE

3.3 Power Functions and Polynomial Functions; 3.4 Graphs of Polynomial Functions; 3.5 Dividing Polynomials; 3.6 Zeros of Polynomial Functions; 3.7 Rational Functions; 3.8 Inverses and Radical Functions; 3.9 Modeling Using Variation; 4.2 Graphs of Exponential Functions; 4.4 Graphs of Logarithmic Functions

M49-PWR-001 — I can identify a power function and interpret the parameters in y = kx^p.

TEXTBOOK SECTIONS 3.3 Power Functions and Polynomial Functions

SUPPORTING SKILLS

D	Interpret the coefficient and exponent in a power function. (first introduced in Math 39) SK-POWER-PARAMETER-INTERPRET · Deepen
D	Represent and interpret a power-variation relationship. (first introduced in Math 39) SK-VAR-POWER · Deepen

M49-PWR-002 — I can construct a power function from two data points or a proportional relationship.

TEXTBOOK SECTIONS 3.9 Modeling Using Variation

SUPPORTING SKILLS

A	Isolate a specified variable in an equation or formula. (first introduced in Math 12) SK-ALG-VARIABLE-ISOLATE · Apply
D	Determine a power function from two points or a proportional relationship. (first introduced in Math 39) SK-POWER-CONSTRUCT · Deepen

M49-PWR-003 — I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.

TEXTBOOK SECTIONS 3.3 Power Functions and Polynomial Functions

SUPPORTING SKILLS

D	Determine power-function behavior as the input approaches zero or positive or negative infinity. (first introduced in Math 39) SK-POWER-END-BEHAVIOR · Deepen
D	Relate the coefficient and exponent of a power function to its graph shape. (first introduced in Math 39) SK-POWER-SHAPE · Deepen
A	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Apply

M49-PWR-004 — I can recognize and model direct, inverse, and other power-variation relationships.

TEXTBOOK SECTIONS 3.9 Modeling Using Variation

SUPPORTING SKILLS

D	Represent and interpret a direct-variation relationship. (first introduced in Math 39) SK-VAR-DIRECT · Deepen
D	Represent and interpret an inverse-variation relationship. (first introduced in Math 39) SK-VAR-INVERSE · Deepen
D	Represent and interpret a power-variation relationship. (first introduced in Math 39) SK-VAR-POWER · Deepen

M49-PWR-005 — I can solve equations involving integer or rational powers.

TEXTBOOK SECTIONS 3.8 Inverses and Radical Functions

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
D	Interpret numerator and denominator roles in a rational exponent. (first introduced in Math 21) SK-EXP-RATIONAL · Deepen

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SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M49-PWR-006

I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.

TEXTBOOK SECTIONS 4.2 Graphs of Exponential Functions; 4.4 Graphs of Logarithmic Functions

SUPPORTING SKILLS

A	Compare functions shown in different representations. (first introduced in Math 21) SK-FUNC-COMPARE-REP · Apply
D	Compare the long-run growth or decay of functions from different families. (first introduced in Math 39) SK-FUNC-DOMINANCE · Deepen

M49-PWR-008

I can analyze power functions with fractional exponents, including their domains, graph shapes, and behavior near zero.

TEXTBOOK SECTIONS 3.8 Inverses and Radical Functions

SUPPORTING SKILLS

A	Interpret numerator and denominator roles in a rational exponent. (first introduced in Math 21) SK-EXP-RATIONAL · Apply
A	Require an even-root radicand to be nonnegative when determining domain. (first introduced in Math 21) SK-FUNC-DOMAIN-EVEN-ROOT · Apply
I	Analyze the domain, graph shape, and behavior near zero of a power function with a fractional exponent. SK-POWER-FRACTIONAL-ANALYZE · Introduce

M49-ALG-002

I can predict the end behavior of a polynomial from its degree and leading coefficient.

TEXTBOOK SECTIONS 3.3 Power Functions and Polynomial Functions; 3.4 Graphs of Polynomial Functions

SUPPORTING SKILLS

I	Infer polynomial end behavior from degree and leading coefficient. SK-POLY-END-BEHAVIOR · Introduce
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

M49-ALG-003

I can determine polynomial zeros and their multiplicities.

TEXTBOOK SECTIONS 3.4 Graphs of Polynomial Functions; 3.5 Dividing Polynomials; 3.6 Zeros of Polynomial Functions

SUPPORTING SKILLS

I	Relate zero multiplicity to local graph behavior. SK-POLY-MULTIPLICITY · Introduce
D	Determine real and complex zeros using an appropriate method. (first introduced in Math 21) SK-POLY-ZEROS · Deepen

M49-ALG-004

I can construct a polynomial function from specified zeros or graphical features.

TEXTBOOK SECTIONS 3.4 Graphs of Polynomial Functions; 3.5 Dividing Polynomials; 3.6 Zeros of Polynomial Functions

SUPPORTING SKILLS

D	Build a polynomial from zeros, multiplicities, and a point. (first introduced in Math 31) SK-POLY-CONSTRUCT · Deepen
A	Relate zero multiplicity to local graph behavior. (first introduced in Math 39) SK-POLY-MULTIPLICITY · Apply
A	Determine real and complex zeros using an appropriate method. (first introduced in Math 21) SK-POLY-ZEROS · Apply

M49-ALG-005

I can sketch and analyze a polynomial function using intercepts, multiplicity, and end behavior.

TEXTBOOK SECTIONS 3.4 Graphs of Polynomial Functions; 3.6 Zeros of Polynomial Functions

SUPPORTING SKILLS

D	Infer polynomial end behavior from degree and leading coefficient. (first introduced in Math 39) SK-POLY-END-BEHAVIOR · Deepen
D	Relate zero multiplicity to local graph behavior. (first introduced in Math 39) SK-POLY-MULTIPLICITY · Deepen
A	Determine real and complex zeros using an appropriate method. (first introduced in Math 21) SK-POLY-ZEROS · Apply

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SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M49-ALG-006 — I can identify holes and vertical, horizontal, or slant asymptotes of a rational function.

TEXTBOOK SECTIONS 3.7 Rational Functions

SUPPORTING SKILLS

A	Cancel common factors rather than individual terms. (first introduced in Math 21) SK-RAT-CANCEL-FACTORS · Apply
I	Determine horizontal or slant asymptotic behavior. SK-RAT-END-BEHAVIOR · Introduce
A	Factor numerators and denominators completely. (first introduced in Math 21) SK-RAT-FACTOR · Apply
I	Identify a removable discontinuity and its coordinates. SK-RAT-HOLE · Introduce
I	Determine vertical asymptotes from noncanceling denominator zeros. SK-RAT-VERTICAL-ASYMPTOTE · Introduce

M49-ALG-007 — I can sketch and analyze a rational function using its algebraic and asymptotic features.

TEXTBOOK SECTIONS 3.7 Rational Functions

SUPPORTING SKILLS

D	Determine horizontal or slant asymptotic behavior. (first introduced in Math 49) SK-RAT-END-BEHAVIOR · Deepen
D	Identify a removable discontinuity and its coordinates. (first introduced in Math 49) SK-RAT-HOLE · Deepen
A	Determine excluded values from an original denominator. (first introduced in Math 21) SK-RAT-RESTRICTIONS · Apply
I	Determine the sign of a rational function on intervals. SK-RAT-SIGN-INTERVAL · Introduce
D	Determine vertical asymptotes from noncanceling denominator zeros. (first introduced in Math 49) SK-RAT-VERTICAL-ASYMPTOTE · Deepen

M49-ALG-008 — I can build or select a power, polynomial, or rational model for a context.

TEXTBOOK SECTIONS 3.9 Modeling Using Variation

SUPPORTING SKILLS

I	Select a function family whose behavior fits a table, graph, or context. SK-MODEL-FAMILY-SELECT · Introduce
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Apply

Limits and Function Comparison

REQUIRED

UNIT TEXTBOOK COVERAGE
12.1 Finding Limits: Numerical and Graphical Approaches

M49-LIM-001 I can interpret limit notation as a statement about function behavior near an input.

TEXTBOOK SECTIONS 12.1 Finding Limits: Numerical and Graphical Approaches

SUPPORTING SKILLS

I	Read and write limit notation with correct approach direction. SK-LIM-NOTATION · Introduce

M49-LIM-002 I can estimate a limit from a graph or table.

TEXTBOOK SECTIONS 12.1 Finding Limits: Numerical and Graphical Approaches

SUPPORTING SKILLS

I	Estimate one-sided and two-sided limits from a graph. SK-LIM-GRAPH · Introduce
I	Estimate a limit from input-output values approaching from both sides. SK-LIM-TABLE · Introduce

M49-LIM-003 I can evaluate a limit algebraically using direct substitution or simplification.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

I	Resolve an indeterminate algebraic form by factoring or rationalizing. SK-LIM-SIMPLIFY · Introduce
I	Test a limit using direct substitution. SK-LIM-SUBSTITUTE · Introduce
A	Factor numerators and denominators completely. (first introduced in Math 21) SK-RAT-FACTOR · Apply

M49-LIM-005 I can interpret an infinite limit as vertical-asymptotic behavior.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

I	Interpret unbounded output near a finite input as an infinite limit. SK-LIM-INFINITE · Introduce
D	Determine vertical asymptotes from noncanceling denominator zeros. (first introduced in Math 49) SK-RAT-VERTICAL-ASYMPTOTE · Deepen

M49-LIM-006 I can determine limits at infinity and relate them to long-run behavior.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

I	Compare dominant terms to determine behavior at infinity. SK-LIM-INFINITY · Introduce
D	Determine horizontal or slant asymptotic behavior. (first introduced in Math 49) SK-RAT-END-BEHAVIOR · Deepen

M49-LIM-007 I can compare the long-run growth of polynomial, exponential, logarithmic, and rational functions.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

A	Compare functions shown in different representations. (first introduced in Math 21) SK-FUNC-COMPARE-REP · Apply
I	Apply the long-run growth hierarchy among logarithmic, polynomial, and exponential functions. SK-LIM-GROWTH-ORDER · Introduce

SKILL PROGRESSION

I Introduce | R Reinforce | D Deepen | A Apply

Sequences and Series

REQUIRED

UNIT TEXTBOOK COVERAGE
11.1 Sequences and Their Notations; 11.2 Arithmetic Sequences; 11.3 Geometric Sequences; 11.4 Series and Their Notations

M49-SEQ-001 I can evaluate and graph a sequence defined explicitly or recursively.

TEXTBOOK SECTIONS 11.1 Sequences and Their Notations

SUPPORTING SKILLS	
I	Generate an explicit formula from a sequence pattern. SK-SEQ-EXPLICIT · Introduce
I	Graph a sequence as discrete ordered pairs. SK-SEQ-GRAPH · Introduce
I	Interpret subscripts and evaluate terms of a sequence. SK-SEQ-NOTATION · Introduce
I	Generate a recursive formula with an initial condition. SK-SEQ-RECURSIVE · Introduce

M49-SEQ-002 I can write explicit and recursive formulas for arithmetic sequences.

TEXTBOOK SECTIONS 11.2 Arithmetic Sequences

SUPPORTING SKILLS	
I	Recognize constant difference in an arithmetic sequence. SK-SEQ-DIFFERENCE · Introduce
D	Generate an explicit formula from a sequence pattern. (first introduced in Math 49) SK-SEQ-EXPLICIT · Deepen
D	Generate a recursive formula with an initial condition. (first introduced in Math 49) SK-SEQ-RECURSIVE · Deepen

M49-SEQ-003 I can write explicit and recursive formulas for geometric sequences.

TEXTBOOK SECTIONS 11.3 Geometric Sequences

SUPPORTING SKILLS	
D	Generate an explicit formula from a sequence pattern. (first introduced in Math 49) SK-SEQ-EXPLICIT · Deepen
I	Recognize constant ratio in a geometric sequence. SK-SEQ-RATIO · Introduce
D	Generate a recursive formula with an initial condition. (first introduced in Math 49) SK-SEQ-RECURSIVE · Deepen

M49-SEQ-004 I can distinguish arithmetic, geometric, and other sequences from formulas, tables, or contexts.

TEXTBOOK SECTIONS 11.1 Sequences and Their Notations; 11.2 Arithmetic Sequences; 11.3 Geometric Sequences

SUPPORTING SKILLS	
A	Compare functions shown in different representations. (first introduced in Math 21) SK-FUNC-COMPARE-REP · Apply
D	Recognize constant difference in an arithmetic sequence. (first introduced in Math 49) SK-SEQ-DIFFERENCE · Deepen
D	Recognize constant ratio in a geometric sequence. (first introduced in Math 49) SK-SEQ-RATIO · Deepen

M49-SEQ-005 I can create and interpret a sequence model for discrete change.

TEXTBOOK SECTIONS 11.2 Arithmetic Sequences; 11.3 Geometric Sequences

SUPPORTING SKILLS	
A	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Apply
A	Generate an explicit formula from a sequence pattern. (first introduced in Math 49) SK-SEQ-EXPLICIT · Apply
I	Identify initial value and pattern of change in a discrete context. SK-SEQ-MODEL · Introduce
A	Generate a recursive formula with an initial condition. (first introduced in Math 49) SK-SEQ-RECURSIVE · Apply

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SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M49-SER-001 I can interpret and use summation notation.

TEXTBOOK SECTIONS 11.4 Series and Their Notations

SUPPORTING SKILLS

A	Interpret subscripts and evaluate terms of a sequence. (first introduced in Math 49) SK-SEQ-NOTATION · Apply
I	Expand and evaluate a finite sum written in sigma notation. SK-SER-SIGMA · Introduce

M49-SER-002 I can calculate a finite arithmetic series.

TEXTBOOK SECTIONS 11.4 Series and Their Notations

SUPPORTING SKILLS

A	Recognize constant difference in an arithmetic sequence. (first introduced in Math 49) SK-SEQ-DIFFERENCE · Apply
I	Apply an arithmetic-series sum formula. SK-SER-ARITHMETIC · Introduce

M49-SER-003 I can calculate a finite geometric series.

TEXTBOOK SECTIONS 11.4 Series and Their Notations

SUPPORTING SKILLS

A	Recognize constant ratio in a geometric sequence. (first introduced in Math 49) SK-SEQ-RATIO · Apply
I	Apply a finite geometric-series sum formula. SK-SER-GEOMETRIC-FINITE · Introduce

M49-SER-004 I can determine whether an infinite geometric series converges and find its sum when it does.

TEXTBOOK SECTIONS 11.4 Series and Their Notations

SUPPORTING SKILLS

A	Recognize constant ratio in a geometric sequence. (first introduced in Math 49) SK-SEQ-RATIO · Apply
I	Test convergence and sum an infinite geometric series. SK-SER-GEOMETRIC-INFINITE · Introduce

M49-SER-005 I can create and interpret a series model for accumulated change.

TEXTBOOK SECTIONS 11.4 Series and Their Notations

SUPPORTING SKILLS

A	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Apply
A	Apply an arithmetic-series sum formula. (first introduced in Math 49) SK-SER-ARITHMETIC · Apply
A	Apply a finite geometric-series sum formula. (first introduced in Math 49) SK-SER-GEOMETRIC-FINITE · Apply
A	Test convergence and sum an infinite geometric series. (first introduced in Math 49) SK-SER-GEOMETRIC-INFINITE · Apply
I	Identify the initial term, change factor, and number of terms in an application. SK-SER-MODEL · Introduce

SKILL PROGRESSION

I	Introduce
R	Reinforce
D	Deepen
A	Apply

Additional Limit Concepts

EXTENSION

UNIT TEXTBOOK COVERAGE

12.1 Finding Limits: Numerical and Graphical Approaches

M49-LIM-004 — I can determine one-sided limits and decide whether a two-sided limit exists.

TEXTBOOK SECTIONS 12.1 Finding Limits: Numerical and Graphical Approaches

SUPPORTING SKILLS

D	Estimate one-sided and two-sided limits from a graph. (first introduced in Math 49) SK-LIM-GRAPH · Deepen
D	Read and write limit notation with correct approach direction. (first introduced in Math 49) SK-LIM-NOTATION · Deepen
A	Estimate a limit from input-output values approaching from both sides. (first introduced in Math 49) SK-LIM-TABLE · Apply

M49-LIM-008 — I can distinguish a function value from a limit and determine continuity at a point.

TEXTBOOK SECTIONS 12.1 Finding Limits: Numerical and Graphical Approaches

SUPPORTING SKILLS

I	Check the three conditions for continuity at a point. SK-LIM-CONTINUITY · Introduce
A	Estimate one-sided and two-sided limits from a graph. (first introduced in Math 49) SK-LIM-GRAPH · Apply
D	Test a limit using direct substitution. (first introduced in Math 49) SK-LIM-SUBSTITUTE · Deepen

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Parametric Equations

EXTENSION

UNIT TEXTBOOK COVERAGE
8.6 Parametric Equations; 8.7 Parametric Equations: Graphs

M49-PAR-001 — I can graph a plane curve defined by parametric equations with its orientation.

TEXTBOOK SECTIONS 8.7 Parametric Equations: Graphs

SUPPORTING SKILLS

A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply
D	Graph a parametric curve with its direction of motion. (first introduced in Math 32) SK-PAR-GRAPH · Deepen
D	Generate ordered pairs from parametric equations. (first introduced in Math 32) SK-PAR-TABLE · Deepen

M49-PAR-002 — I can eliminate a parameter to obtain a rectangular equation.

TEXTBOOK SECTIONS 8.6 Parametric Equations

SUPPORTING SKILLS

A	Isolate a specified variable in an equation or formula. (first introduced in Math 12) SK-ALG-VARIABLE-ISOLATE · Apply
D	Eliminate a parameter algebraically. (first introduced in Math 32) SK-PAR-ELIMINATE · Deepen
A	Substitute an equivalent expression for a variable. (first introduced in Math 12) SK-SYS-SUBSTITUTE · Apply

M49-PAR-003 — I can create or interpret parametric equations for motion in the plane.

TEXTBOOK SECTIONS 8.6 Parametric Equations; 8.7 Parametric Equations: Graphs

SUPPORTING SKILLS

D	Interpret or construct parametric equations for planar motion. (first introduced in Math 32) SK-PAR-MODEL · Deepen
A	Generate ordered pairs from parametric equations. (first introduced in Math 32) SK-PAR-TABLE · Apply
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Conic Sections

EXTENSION

UNIT TEXTBOOK COVERAGE

10.1 The Ellipse; 10.2 The Hyperbola; 10.3 The Parabola; 10.5 Conic Sections in Polar Coordinates

M49-CON-001 I can classify a circle, ellipse, or hyperbola from its equation or defining geometric relationship.

TEXTBOOK SECTIONS 10.1 The Ellipse; 10.2 The Hyperbola; 10.5 Conic Sections in Polar Coordinates

SUPPORTING SKILLS	
I	Classify a circle, ellipse, hyperbola, or parabola from an equation or geometric definition. SK-CONIC-CLASSIFY · Introduce
I	Use fixed-distance and focal-distance relationships to explain circles, ellipses, and hyperbolas. SK-CONIC-DISTANCE-DEFINITION · Introduce

M49-CON-002 I can use completing the square to rewrite a general-form conic equation in standard form.

TEXTBOOK SECTIONS 10.1 The Ellipse; 10.2 The Hyperbola; 10.3 The Parabola

SUPPORTING SKILLS	
A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Complete squares to convert a general conic equation to standard form. SK-CONIC-GENERAL-STANDARD · Introduce
D	Create a perfect-square trinomial by completing the square. (first introduced in Math 21) SK-QUAD-COMPLETE-SQUARE · Deepen

M49-CON-003 I can write, interpret, and graph implicit and parametric equations of a circle.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS	
I	Connect implicit, standard, and parametric equations of a circle. SK-CONIC-CIRCLE-EQUATIONS · Introduce
I	Determine and use a circle's center and radius. SK-CONIC-CIRCLE-FEATURES · Introduce
A	Graph a parametric curve with its direction of motion. (first introduced in Math 32) SK-PAR-GRAPH · Apply
A	Generate ordered pairs from parametric equations. (first introduced in Math 32) SK-PAR-TABLE · Apply

M49-CON-004 I can write, interpret, and graph implicit and parametric equations of an ellipse.

TEXTBOOK SECTIONS 10.1 The Ellipse

SUPPORTING SKILLS	
I	Connect implicit, standard, and parametric equations of an ellipse. SK-CONIC-ELLIPSE-EQUATIONS · Introduce
I	Determine and use an ellipse's center, axes, vertices, and foci. SK-CONIC-ELLIPSE-FEATURES · Introduce
A	Graph a parametric curve with its direction of motion. (first introduced in Math 32) SK-PAR-GRAPH · Apply
A	Generate ordered pairs from parametric equations. (first introduced in Math 32) SK-PAR-TABLE · Apply

Math 49 In-Depth Curriculum

Precalculus with Limits

TEXTBOOK | OpenStax Precalculus 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M49-CON-005 — I can write, interpret, and graph implicit and parametric equations of a hyperbola.

TEXTBOOK SECTIONS 10.2 The Hyperbola

SUPPORTING SKILLS

I	Connect implicit, standard, and parametric equations of a hyperbola. SK-CONIC-HYPERBOLA-EQUATIONS · Introduce
I	Determine and use a hyperbola's center, axes, vertices, foci, and asymptotes. SK-CONIC-HYPERBOLA-FEATURES · Introduce
A	Graph a parametric curve with its direction of motion. (first introduced in Math 32) SK-PAR-GRAPH · Apply
A	Generate ordered pairs from parametric equations. (first introduced in Math 32) SK-PAR-TABLE · Apply

M49-CON-006 — I can determine the center, vertices, foci, axes, or asymptotes needed to describe a circle, ellipse, or hyperbola.

TEXTBOOK SECTIONS 10.1 The Ellipse; 10.2 The Hyperbola; 10.5 Conic Sections in Polar Coordinates

SUPPORTING SKILLS

D	Determine and use a circle's center and radius. (first introduced in Math 49) SK-CONIC-CIRCLE-FEATURES · Deepen
A	Classify a circle, ellipse, hyperbola, or parabola from an equation or geometric definition. (first introduced in Math 49) SK-CONIC-CLASSIFY · Apply
D	Determine and use an ellipse's center, axes, vertices, and foci. (first introduced in Math 49) SK-CONIC-ELLIPSE-FEATURES · Deepen
D	Determine and use a hyperbola's center, axes, vertices, foci, and asymptotes. (first introduced in Math 49) SK-CONIC-HYPERBOLA-FEATURES · Deepen

M49-CON-007 — I can explain the geometric definitions of circles, ellipses, and hyperbolas using foci and distance relationships.

TEXTBOOK SECTIONS 10.1 The Ellipse; 10.2 The Hyperbola; 10.5 Conic Sections in Polar Coordinates

SUPPORTING SKILLS

D	Use fixed-distance and focal-distance relationships to explain circles, ellipses, and hyperbolas. (first introduced in Math 49) SK-CONIC-DISTANCE-DEFINITION · Deepen
A	Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula. (first introduced in Math 22) SK-COORD-DISTANCE-PYTHAGOREAN · Apply

M49-CON-008 — I can explain a parabola as the set of points equidistant from a focus and directrix.

TEXTBOOK SECTIONS 10.3 The Parabola; 10.5 Conic Sections in Polar Coordinates

SUPPORTING SKILLS

I	Explain a parabola using its focus-directrix distance relationship. SK-CONIC-PARABOLA-DEFINITION · Introduce
A	Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula. (first introduced in Math 22) SK-COORD-DISTANCE-PYTHAGOREAN · Apply